On Stochastic Sign Descent Methods

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The Problem

$$\min_{x\in\mathbb{R}^d} f(x)$$

- f is lower bounded: $f(x) \geq f^*$ for any $x \in \mathbb{R}^d$
- f is $L-$smooth with some non-negative constants $L = [L_1, \ldots, L_d]$

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \tfrac{1}{2}\sum_{i=1}^{d} L_i (x_i - y_i)^2, \; x, y \in \mathbb{R}^d$$

- f can be non-convex

The Algorithm

Algorithm 1 SIGNSGD

1: **Input:** step size γ_k, current point x_k
2: $\hat{g}_k \leftarrow \text{StochasticGradient}(x_k)$
3: $x_{k+1} \leftarrow x_k - \gamma_k \,\text{sign}\,\hat{g}_k$

The Assumption

Success Probability Bounds (SPB). For any $x \in \mathbb{R}^d$, we have access to an independent (and not necessarily unbiased) estimator $\hat{g}(x)$ of the true gradient $g(x) \coloneqq \nabla f(x)$ that satisfies

$$\rho_i(x) \coloneqq \Pr\left(\text{sign}\,\hat{g}_i(x) = \text{sign}\,g_i(x)\right) > \tfrac{1}{2}, \quad \text{if} \quad g_i(x) \neq 0 \tag{1}$$

for all $x \in \mathbb{R}^d$ and all $i \in \{1, 2, \ldots, d\}$.

A New "Norm" for Measuring Gradients

Definition 1 (ρ-norm). Let $\rho \coloneqq \{\rho_i(x)\}_{i=1}^d$ be the collection of probability functions from the SPB assumption. Define the ρ-norm of gradient $g(x)$ via

$$\|g(x)\|_\rho \coloneqq \sum_{i=1}^{d} (2\rho_i(x) - 1)|g_i(x)|.$$

Although *ρ-norm is not a norm*, it is always *positive definite*, i.e.

$$\|g\|_\rho \geq 0, \text{ and } \|g\|_\rho = 0 \text{ if and only if } g = 0.$$

Under unimodal and symmetric noise assumption, ρ-norm can be lower bounded by a mixture of the l^1 and squared l^2 norms:

$$\|g\|_\rho = \sum_{i=1}^{d}(2\rho_i - 1)|g_i| \geq \sum_{i=1}^{d} \frac{g_i^2}{|g_i| + \sqrt{3}\sigma_i} \coloneqq \|g\|_{l^{1,2}}. \tag{2}$$

Note that $l^{1,2}$-norm is again not a norm. However, it is positive definite, continuous and order preserving, i.e., for any g^k, g, $\tilde{g} \in \mathbb{R}^d$ we have

- $\|g\|_{l^{1,2}} \geq 0$ and $\|g\|_{l^{1,2}} = 0$ if and only if $g = 0$,
- $g^k \to g$ (in l^2 sense) implies $\|g^k\|_{l^{1,2}} \to \|g\|_{l^{1,2}}$,
- $\|g^k\|_{l^{1,2}} \to 0$ implies $g^k \to 0$,
- $0 \leq g_i \leq \tilde{g}_i$ for any $1 \leq i \leq d$ implies $\|g\|_{l^{1,2}} \leq \|\tilde{g}\|_{l^{1,2}}$.

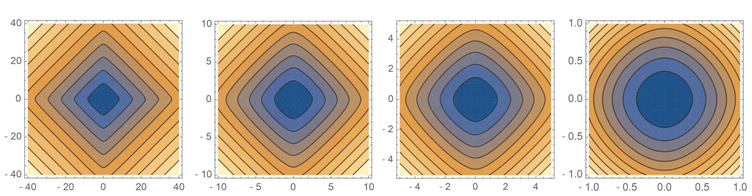

Figure 1: Contour plots of the $l^{1,2}$ norm (2) at 4 different scales with fixed noise $\sigma = 1$.

Main Contributions

- The key novelty of our methods is in a *substantial relaxation* of the requirements on the gradient estimator $\hat{g}(x_k)$ of the true gradient $\nabla f(x^k)$. In sharp contrast with existing approaches [1,2], we allow $\hat{g}(x_k)$ to be *biased*. We show through a counterexample that SPB assumption is *necessary* [3].
- As a byproduct of our analysis, we uncover a *mixed l^1-l^2 geometry* of sign descent methods.
- While our complexity bounds have the same $\mathcal{O}(1/\sqrt{K})$ dependence on the number of iterations, they have a *better dependence on the smoothness parameters* associated with f. We present first result on signSGD for non-convex functions which *does not rely on mini-batching*, and which allows for *step sizes independent of the total number of iterations K*. Our bounds are cast in terms of a *novel norm-like function*, which we call the *ρ-norm*, which is a weighted l^1 norm with positive variable weights.
- We extend our results to the *distributed setting* with arbitrary # of nodes, where we also consider sign-based compression of the aggregated gradients.

Sufficient conditions for SPB

Suppose we have access to an unbiased estimator $\hat{g}(x)$ of true gradient $g(x)$.

Lemma 1. If each coordinate $\hat{g}_i$ has a unimodal and symmetric distribution with variance $\sigma_i^2 = \sigma_i^2(x)$, $1 \leq i \leq d$. Then

$$\rho_i \geq \tfrac{1}{2} + \tfrac{1}{2}\tfrac{|g_i|}{|g_i| + \sqrt{3}\sigma_i} > \tfrac{1}{2} \quad \text{if} \quad g_i \neq 0.$$

Lemma 2. If coordinate-wise variances $\sigma_i^2(x) \leq c\, g_i^2(x)$ are bounded for some constant $C \geq 0$. Then, choosing mini-batch size $\tau > 2c$, we have

$$\rho_i \geq 1 - \tfrac{c}{\tau} > \tfrac{1}{2}, \quad \text{if} \quad g_i \neq 0.$$

Lemma 3. Let $\sigma_i^2 = \sigma_i^2(x)$ be the variance and $\nu_i^3 = \nu_i^3(x)$ be the 3th central moment of $\hat{g}_i(x)$, $1 \leq i \leq d$. Then

$$\rho_i > \tfrac{1}{2} \quad \text{if mini-batch size } \tau > 2\min\left(\tfrac{\sigma_i^2}{g_i^2}, \tfrac{\nu_i^3}{|g_i|\sigma_i^2}\right).$$

A counterexample to signSGD [3]

$$\min_{x\in\mathbb{R}^2} f(x) = \tfrac{1}{2}\left[\langle a_1, x\rangle^2 + \langle a_2, x\rangle^2\right], \quad a_1 = \left[\begin{smallmatrix}1+\varepsilon\\ -1+\varepsilon\end{smallmatrix}\right], a_2 = \left[\begin{smallmatrix}-1+\varepsilon\\ 1+\varepsilon\end{smallmatrix}\right],$$

where $\varepsilon \in (0,1)$ and $\hat{g}(x) = \nabla\langle a_i, x\rangle^2 = 2\langle a_i, x\rangle a_i$ with probs $1/2$. Then

$$\text{sign}\,\hat{g}(x) = (-1)^i \,\text{sign}\langle a_i, x\rangle \left[\begin{smallmatrix}-1\\ 1\end{smallmatrix}\right] \quad \text{with probabilities} \quad 1/2.$$

Notice that signSGD with any step-size remains stuck since

$$x \in H := \{(z_1, z_2) \colon z_1 + z_2 = 2\} \implies \text{sign}\,\hat{g}(x) = \pm(1, -1) \parallel H.$$

Here, SPB assumption (1) is violated as $\rho_i(x) \leq 1/2$ when $\langle a_1, x\rangle \cdot \langle a_2, x\rangle > 0$.

Convergence Analysis

Non-convex convergence of signSGD

Under the SPB assumption, signSGD (Algorithm 1) with step sizes $\gamma_k = \gamma_0/\sqrt{k+1}$ converges as follows

$$\min_{0\leq k<K} \mathbb{E}\|\nabla f(x_k)\|_\rho \leq \tfrac{1}{\sqrt{K}}\left[\tfrac{f(x_0)-f^*}{\gamma_0} + \gamma_0 d\bar{L}\right] + \tfrac{\gamma_0 d\bar{L}}{2}\tfrac{\log K}{\sqrt{K}}.$$

If $\gamma_k \equiv \gamma > 0$, we get $1/K$ convergence to a neighbourhood of the solution:

$$\tfrac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(x_k)\|_\rho \leq \tfrac{f(x_0)-f^*}{\gamma K} + \tfrac{d\bar{L}}{2}\gamma.$$

Distributed Setting

Algorithm 2 DISTRIBUTED SIGNSGD WITH MAJORITY VOTE

1: **Input:** step size γ_k, current point x_k, # of nodes M
2: **on** each node
3: $\quad\hat{g}^m(x_k) \leftarrow \text{StochasticGradient}(x_k)$
4: **on** server
5: $\quad$**pull** $\text{sign}\hat{g}^m(x_k)$ **from** each node
6: $\quad$**push** $\text{sign}\left[\sum_{m=1}^M \text{sign}\hat{g}^m(x_k)\right]$ **to** each node
7: **on** each node
8: $\quad x_{k+1} \leftarrow x_k - \gamma_k \text{sign}\left[\sum_{m=1}^M \text{sign}\hat{g}^m(x_k)\right]$

Definition 2 (ρ_M-norm). Let $l = \left[\frac{M+1}{2}\right]$ and I is the regularized incomplete beta function. Define ρ_M-norm of gradient $g(x)$ at $x \in \mathbb{R}^d$ as

$$\|g(x)\|_{\rho_M} = \sum_{i=1}^{d} \left(2I(\rho_i(x); l, l) - 1\right)|g_i(x)|$$

Non-convex convergence of distributed signSGD

Under SPB assumption, distributed signSGD (Algorithm 2) with step sizes $\gamma_k = \gamma_0/\sqrt{k+1}$ converges as follows

$$\min_{0\leq k<K} \mathbb{E}\|\nabla f(x_k)\|_{\rho_M} \leq \tfrac{1}{\sqrt{K}}\left[\tfrac{f(x_0)-f^*}{\gamma_0} + \gamma_0 d\bar{L}\right] + \tfrac{\gamma_0 d\bar{L}}{2}\tfrac{\log K}{\sqrt{K}}.$$

If $\gamma_k \equiv \gamma > 0$, we have convergence up to a level proportional to γ:

$$\tfrac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\|\nabla f(x_k)\|_{\rho_M} \leq \tfrac{f(x_0)-f^*}{\gamma K} + \tfrac{d\bar{L}}{2}\gamma.$$

- **Exponentially fast variance reduction.**

$$\left(1 - \exp\left(-(2\rho(x) - 1)^2 l\right)\right)\|g(x)\|_1 \leq \|g(x)\|_{\rho_M} \leq \|g(x)\|_1$$

where $\rho(x) = \min_{1\leq i\leq d} \rho_i(x) > 1/2$.

- **No theoretically difference between $2l - 1$ and $2l$ nodes.**

$$\mathbb{E}\left[\text{sign}(\hat{g}_i^{(2l)} \cdot g_i)\right] = \mathbb{E}\left[\text{sign}(\hat{g}_i^{(2l-1)} \cdot g_i)\right]$$

where $\hat{g}^{(M)}$ is the sum of stochastic sign vectors aggregated from nodes.

Experiments

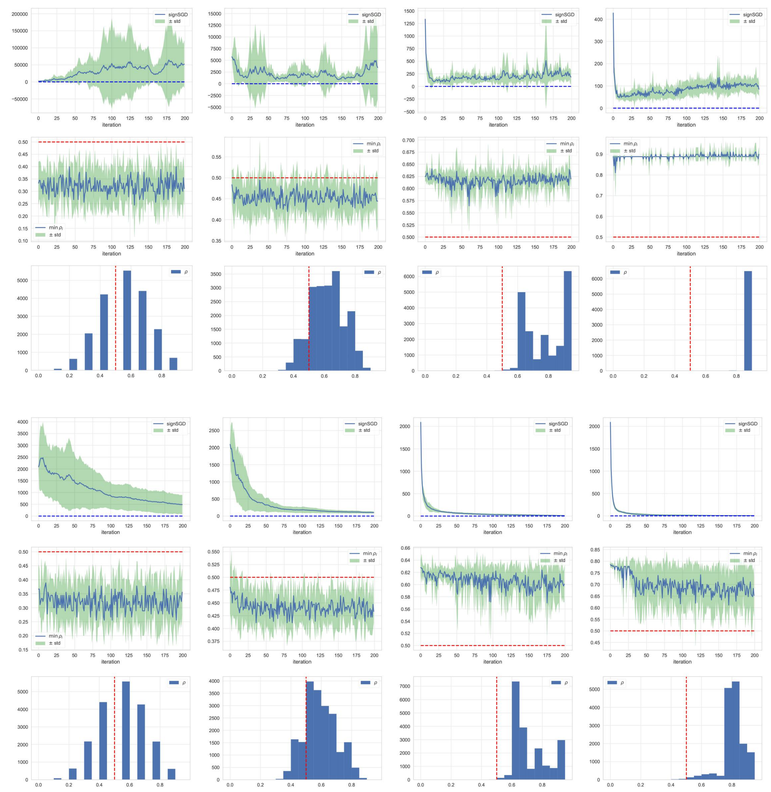

Figure 2: Performance of signSGD with constant (*Top 3 rows*) and decreasing (*Botton 3 rows*) step sizes under four different noise levels using the Rosenbrock function.

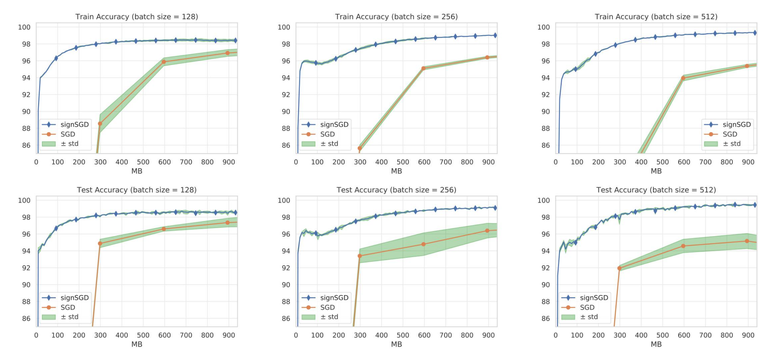

Figure 3: Comparison of signSGD and SGD on MNIST dataset with a fixed budget of gradient communication (MB) using single hidden layer feed-forward neural network.

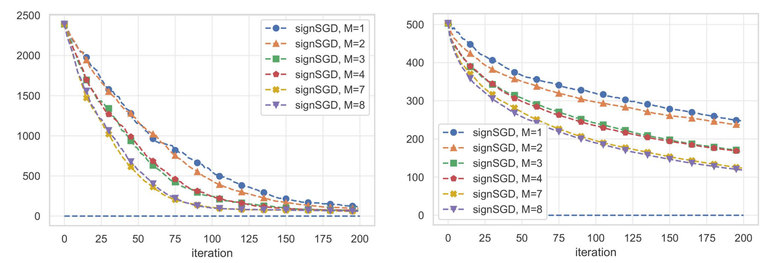

Figure 4: Experiments on distributed signSGD with constant (*Left*) and decreasing (*Right*) step size regimes using the Rosenbrock function.

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